D1 - PROJECT PROPOSAL

AgriUAV Platform: Precision Drone Dispatching Platform in the Mekong Delta

Subject	ITA301 - Information System Analysis & Design
Deliverable	D1 - Project Proposal (15%)
Submission Date	2026-06-25
Version	v1.6
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1. Problem Statement

Agricultural cooperatives and drone service enterprises in the Mekong Delta currently face a critical lack of digital tools to schedule, dispatch, and monitor spraying drone fleets, resulting in the following measurable consequences:

High Operational Costs: Managing spraying schedules manually via Zalo, Excel, or paper notebooks causes schedule conflicts, overlapping shifts, and wasted deadhead flights, inflating operational costs by an estimated 15-20%[1].

Low Field-Level Efficiency: Without accurate field mapping, drones spray uniformly across the entire field rather than concentrating variable dosages on specific pest-infected zones, resulting in excessive agrochemical wastage.

Safety Risks & Heavy Losses: Manual operations provide no shared hazard visibility or early warning when a drone drifts off-course toward neighbouring plots or registered high-voltage corridors. The Long An incident, in which an agricultural drone contacted a 110kV transmission line and caused blackouts for 76,000 customers, illustrates why hazard visibility, early warning, incident logging, and operational traceability are needed (Tuoi Tre, 2024).

Compliance Hardships: No automated flight logging tools exist, failing to satisfy mandatory reporting requirements regarding drone identification numbers and pilot certifications specified under Decree 288/2025/ND-CP (effective November 5, 2025).

1.1. Proposed Solution

The AgriUAV Platform is designed as a Web and Mobile application ecosystem that supports cooperative scheduling, recommends rule-based drone–pilot assignments via digital field mapping, supports formula-based pesticide dosage calculations for authorized confirmation, and tracks spraying progress in near real-time. Within the ITA301 prototype scope, location and operational data are represented through operator mobile GPS or structured simulated telemetry rather than direct integration with physical drone firmware.

1.2. System Scope

Operation management of agricultural spraying drone fleets within the rice cultivation regions of the Mekong Delta. For the current ITA301 design and evaluation, the reference workload is one cooperative managing up to 50 concurrently monitored drone telemetry streams. The registered fleet size is configurable per cooperative and is not implemented as a permanent hard-coded limit; beyond 50 concurrent drones the system does not automatically prohibit usage, but the latency, reliability, and performance NFRs have not yet been confirmed within the current study scope.

1.3. In-Scope - System Functions

- Drone fleet management: drone identification codes, model specifications, tank capacity, and operational status (Available / In-Flight / Under Maintenance)
- Pilot profile management: certifications, licenses, and flight history records aligned with the reporting requirements of Decree 288/2025/ND-CP
- Spray scheduling: farmers and HTX managers submit spray requests; system confirms and records assignments
- Rule-based drone dispatch recommendation: the system suggests suitable drone–pilot assignments using availability, battery status, GPS position, maintenance condition, pilot certification validity, and schedule-conflict checks; the HTX Manager reviews and confirms the final assignment

[1] The 15-20% figure is a team estimate (a working planning assumption), not a measured or published value. Nguyen et al. (2025) is a Probit analysis of the determinants of UAV adoption among 940 Mekong Delta farm households (RWAE Journal); the only figures this report draws from that study are its sample size (N=940) and the average farmer age (~50.88 years, Table 1). The study does not measure or report an operational-cost-inflation rate. The 15-20% reflects the team's reasoning about deadhead flights, double-booking, and overlapping shifts under manual Zalo/Excel scheduling, and is treated as an assumption subject to field validation (see Section 2.3).

- Near real-time spray progress tracking on digital field map: drone position overlay and percentage area completion, fed by operator mobile GPS or structured simulated telemetry within the prototype scope
- Automated alerts: low pesticide level, low battery, adverse weather (wind speed ≥ 3 m/s, an assumed threshold pending expert confirmation), and proximity to or crossing of a configured geofence boundary (including a 50 m buffer zone around registered high-voltage power lines)
- Pesticide inventory management and formula-based dosage support per approved chemical registry, with the final formula and dosage requiring confirmation by an authorized agronomist or HTX Manager before mission dispatch
- Spray log export and seasonal compliance reporting for the Plant Protection Department
- Role-based access control: Admin / HTX Manager / Operator / Farmer / Agronomist
- System scale: validation baseline of one cooperative managing up to 50 concurrently monitored drone telemetry streams within rice cultivation zones in the Mekong Delta; the registered fleet size is configurable per cooperative and is not a hard-coded limit

1.4. Out-of-Scope - Explicitly Excluded

- Direct drone hardware control from software (fly-by-wire or autopilot command interfaces)
- AI-based pest detection from drone imagery (computer vision or deep learning models)
- Integration with irrigation systems or other agricultural machinery
- Financial management, accounting, or VAT invoicing modules
- Online flight permit application with the Operations Department (UTM system integration)
- Physical cloud deployment or real-hardware drone integration

2. Business Case & ROI Analysis

2.1. Ecosystem Analysis - 3 Layer Model

To examine the market gap that the AgriUAV Platform is intended to address, the domain structure is categorized into three explicit layers:

Ecosystem Layer	Current Market State	AgriUAV Platform Intervention
Hardware Layer	Highly developed and saturated - DJI Agras (~80% Vietnam market share), XAG (~15%), YANMAR, and domestic brands (~5%), equipped with RTK positioning and multispectral cameras. Hardware availability is not a bottleneck.	Agnostic integration via mobile GPS telemetry at MVP stage. Planned native MAVLink telemetry protocol integration in future versions.
Service Layer	Local cooperatives (HTX) and private entities provide bulk drone spraying services (~150,000 VND/ha). Internal operations remain largely manual and uncoordinated - managed via Zalo, Excel, and paper logbooks.	Provides the end-to-end digital coordination layer to support scheduling, rule-based dispatch recommendations, and compliance record-keeping for existing service providers.
Platform Layer	The reviewed evidence indicates a lack of a widely documented, integrated, and localized information platform that combines cooperative-level scheduling, dispatching, telemetry monitoring, incident management, and compliance-record support for agricultural drone services in the Mekong Delta. Thanh Nien (June 2025) reports 1,215 active drones across 3 provinces, with no platform-layer management software described in the source.	AgriUAV Platform is designed to address this gap as an operational coordination system for drone service cooperatives.

2.2. Market Sizing (TAM / SAM / SOM)

Market sizing draws on reported field statistics as of June 2025 and empirical research. The SAM is based on verified provincial drone counts, whereas the TAM is an approximate regional estimate and the SOM is a projected planning figure. Note: the SOM figure is a projected estimate based on the Can Tho University - HTX extension partnership network; it is not a signed commitment and should be treated as a planning assumption.

Segment	Drone Count & Data Source	Target Customer	Strategic Rationale
TAM **(Total Addressable Market)**	>3,000 agricultural drones currently operating, serving approximately 1.5 million ha of rice cultivation in the Mekong Delta. *Source: VnExpress (2025), citing Markets & Data*	Full Mekong Delta drone fleet	The broader regional fleet represents a potential addressable market, subject to customer validation and competitive analysis
SAM	1,215 active drones in 3 priority provinces: Kien Giang (690), An Giang (265), Dong Thap (260).	HTX and commercial drone service operators in 3 core provinces	Operationally reachable in Year 1 rollout

(Serviceable Addressable Market)	*Source: Thanh Nien Newspaper, June 2025 (verified field statistics)*		
SOM **(Serviceable Obtainable Market)**	200 drones managed by pilot-partner cooperatives linked to Can Tho University agricultural research network. *Note: this is a projected SOM based on the Can Tho University-HTX extension partnership network; not yet a signed contract. Conservative estimate subject to revision.*	Strategic pilot HTX partners	Direct partnership channel for initial deployment

Revenue Model: B2B SaaS licensing at 500,000 VND / drone / month. Projected annual recurring revenue from SOM: 200 × 500,000 × 12 = **1,200,000,000 VND/year.**

2.3. Cost-Benefit Analysis (Reference: 1 HTX - 10 drones, 500 ha/season)

The following cost-benefit breakdown uses a representative mid-size cooperative as the reference unit. All AS-IS figures are team estimates derived from published operational benchmarks and a team-estimated 15–20% cost-inflation rate (a planning assumption, not a figure reported in Nguyen et al. (2025), which is a UAV-adoption study; see the note in Section 1). Absolute VND values are conservative projections, not direct field measurements. The 30% pesticide-reduction, 20% fuel-reduction, and 75% administrative-labour-reduction figures are scenario assumptions used for planning; they are not results measured from a prototype or a field pilot. Methodology notes are included in the rightmost column.

Cost Category	AS-IS Manual	TO-BE AgriUAV	Savings	Data Source / Methodology Note
Pesticide wastage (imprecise spraying)	~45M VND/season	~31.5M VND/season	–30% = 13.5M VND	*Derived from DJI Agras T40 average application rate (11 L/ha) vs. precision dosing, applied to 500 ha × 150,000 VND/ha service cost benchmark. Conservative estimate; team derivation based on published operational benchmarks (DJI Agras T40 application data).*
Drone fuel (duplicate flights, wrong zones)	~12M VND/season	~9.6M VND/season	–20% = 2.4M VND	*Based on the team-estimated 15–20% deadhead-flight waste rate (team assumption, not reported in Nguyen et al.); applied to estimated 10-drone fleet fuel cost at*

				~1.2M VND/drone/season.
Coordination & reporting labor	~8M VND/season	~2M VND/season	–75% = 6.0M VND	*Estimated 4 admin-hours/week × 12 weeks × ~170,000 VND/hour = ~8.2M VND AS-IS. The scenario assumes a 75% reduction in manual administrative effort through scheduling and report automation. Team estimate; no direct external source.*
Legal penalty risk (Decree 288 non-compliance)	Unquantified - ongoing exposure	Significantly reduced	Compliance-support functions	*Decree 288/2025/ND-CP includes administrative requirements for compliant flight logging; specific penalty provisions are not confirmed in the reviewed sources, so this risk is treated qualitatively.*
TOTAL ESTIMATED SAVINGS / SEASON	**—**	**—**	**~21.9M VND**	***Sum of above quantified rows.***

From the cooperative's perspective, the subscription fee for the 10-drone reference unit is 10 × 500,000 = 5,000,000 VND/month, i.e. 15,000,000 VND for a three-month rice season. Against the estimated gross seasonal savings of ~21,900,000 VND, the net estimated benefit is ~6,900,000 VND, giving a preliminary ROI of (21.9M − 15M) / 15M ≈ 46%. The separate provider-side infrastructure and API operating cost of ~1,200,000 VND/month (~3,600,000 VND/season) is borne by the platform operator and is not used in the cooperative ROI calculation. These absolute VND values are team estimates and scenario assumptions, not measurements taken at a specific cooperative. Preliminary scenario outcome: the scenario suggests a potentially positive first-season ROI, subject to validation with actual cooperative operating data.

Key performance improvements (summary):

- Operational cost reduction: the team-estimated 15 - 20% manual-cost inflation rate (a planning assumption, not a figure from Nguyen et al. (2025); see the note in Section 1) implies a corresponding reduction as an expected outcome subject to field validation
- Compliance report turnaround: from an estimated 3 - 5 business days (manual) to a design target of under 1 hour (automated PDF/Excel export)
- Hazard awareness: information-level alerts notify the operator and HTX Manager when a drone approaches a registered hazard zone such as a high-voltage transmission line corridor
- Operator activity verification: GPS-based field check-in/out records evidence supporting presence verification for each mission

3. Stakeholder Analysis

The system serves stakeholders across three tiers. Two actors marked (*) were supplemented by the team after identifying gaps in the initial AI-generated stakeholder list, verified against Decree 288/2025/ND-CP and cooperative supply chain domain knowledge. Not all stakeholders are authenticated system users: the direct system actors are the Farmer, Operator/Drone Pilot, HTX Manager, Admin, and Agronomist, whereas the Plant Protection Department, Operations Department, and provincial authorities are report recipients or regulatory stakeholders. Indirect or future stakeholders such as banks, agricultural exporters, pesticide retailers, and drone manufacturers are not MVP users; their integrations (bank data exchange, exporter integration, pesticide-retailer inventory API, and MAVLink manufacturer integration) are treated as post-MVP future work.

Classification	Stakeholder	Core Role & Expectations	Power	Interest	Strategic System Action
Primary	**Cooperative Management (HTX)**	Optimize fleet dispatching, lower operational costs by an estimated 15–20%, reduce unverified activity, and improve the traceability of pesticide usage.	5	5	Intuitive Web Dashboard for scheduling, shift allocation, and compliance reporting.
Primary	**Operators / Drone Pilots**	Receive clear flight orders, view digital field boundaries, navigate flight paths safely away from power lines, and log verified flight hours.	2	5	Lightweight Mobile App with GPS navigation and automated field check-in/check-out logging.
Primary	**Rice Farmers**	Easily book a spray service, track near real-time drone completion status, and verify actual pesticide usage. Avg. age ~51 (Nguyen et al. 2025), limited IT experience.	3	4	Automated SMS/Zalo notifications updating order status as it changes.
Primary	**Agricultural Engineers / Agronomists**	Define and validate spray formulas per crop type, manage approved pesticide registry, and confirm correct dosage application.	3	4	Formula management module with dosage confirmation workflow; Agronomist role in RBAC grants access to configure spray formulas and confirm dosages before dispatch (FR18).
Secondary	**Agricultural Exporters (Trung An, Minh Phu)**	Trace pesticide application history data to meet strict clean grain export standards (EU, US) - VietGAP/GlobalGAP.	3	4	Exportable flight log modules matching VietGAP/GlobalGAP data standards (exporter integration planned post-MVP).

Secondary	**Drone Manufacturers (DJI, XAG)**	Expand market reach through platform compatibility; benefit from telemetry integration via MAVLink SDK.	2	3	MAVLink telemetry API integration roadmap (planned post-MVP).
Secondary	**Banks / Agricultural Credit Institutions**	Use verified seasonal production data as a basis for agricultural lending risk assessment.	2	2	Seasonal operation summary export with verified flight logs usable as credit evidence (bank data exchange planned post-MVP).
Secondary (*)	**Pesticide Retailers**	Supply approved chemical inputs to HTX; inventory sync prevents unapproved pesticide use. [Supplemented by team - not in initial AI output]	2	2	Chemical inventory sync API connected to approved chemical registry (retailer integration planned post-MVP).
Regulatory	**Plant Protection Dept. (Cục Trồng trọt & BVTV)**	Monitor compliance and check approved pesticide databases under Decree 288/2025.	5	3	Automated report generation (PDF/Excel) linking drone serial numbers and pilot licenses.
Regulatory (*)	**Operations Dept. / General Staff (Cục Tác chiến)**	Receives and processes UAV flight-permit dossiers (routed to Cục Tác chiến — confirmed current practice). Flight-permit authority itself is assigned by Decree 288/2025/ND-CP (Điều 21, Khoản 2) to the Ministry of National Defence and Ministry of Public Security. [Supplemented by team; dossier-routing role confirmed via Vietnam Naval News, 2025]	4	2	Store and display device tracking UUIDs and certified pilot license numbers in compliance reports.
Regulatory	**Provincial Authorities (UBND / Sở NN&PTNT ĐBSCL)**	Grant operational licenses for drone service providers at provincial level; receive periodic area-sprayed reports.	3	3	Periodic provincial spraying area reports exportable to Ministry of Agriculture.

4. AS-IS Process Analysis

The current manual operational workflow is broken down into 5 sequential phases. Field evidence and documented incidents expose critical pain points at each stage that require digital infrastructure solutions.

#	Phase	AS-Is Manual Workflow	Measurable Pain Points
1	**Demand Capture**	Farmers call or text the cooperative via Zalo to declare field areas needing spraying.	Ambiguous Boundaries: Lack of GPS coordinates leads to spraying wrong fields, causing accidental neighbor crop exposure and boundary disputes.
2	**Scheduling & Assignment**	Management writes details in a paper journal or logs them in a shared Excel file, then calls pilots individually.	Double-Booking & Deadhead Flights: Scheduling conflicts cause overlapping shifts and misallocated dispatch routes, inflating operational costs by an estimated 15–20% (team estimate; see the note in Section 1).
3	**Field Execution**	Pilots bring the drone to the field and spray based on intuition or standalone hardware controls with no centralized telemetry.	Unverified Records Risk: Manual operation creates a risk of inaccurate, incomplete, or unverifiable flight-hour and chemical-usage reporting, because no centralized audit trail exists.
4	**Monitoring & Control**	No active tracking infrastructure exists. Management waits at the office until the pilot verbally confirms completion by phone.	Safety Visibility Gap: Without shared hazard visibility or early warning, drones may approach high-voltage power lines or drift off-course toward neighbouring plots. The Long An 110kV incident (Tuoi Tre 2024 - 76,000 customers affected) illustrates the severity of such events and the need for hazard visibility, early warning, and incident logging.
5	**Reconciliation & Reporting**	At month-end, the cooperative tallies paper logs to invoice farmers and generates handwritten government reports.	Compliance Failure: Error-prone manual data cannot reliably meet the drone-ID and pilot-credential logging requirements of Decree 288/2025/ND-CP. No verifiable audit trail exists.

5. Work Breakdown Structure (WBS)

WBS	Deliverable / Task	Member	Week	Specific Output
1.0	**D1 - Project Proposal**	**All**	**W1-W3**	**Complete proposal file**
1.1	Choose topic, analyze the UAV domain	BA + PM	W1	Preliminary list of groups and topics
1.2	Stakeholder mapping + AS-IS analysis	BA	W2	Stakeholder map + Business case draft
1.3	Business case, cost-benefit, ROI	BA + PM	W3	Business case with specific data
1.4	WBS + Timeline + Risk Register + Gantt Chart	PM	W3	WBS table, Gantt chart (Section 6.1), risk log
1.5	Internal peer review + completion of D1	All	W3	Submit D1
2.0	**D2 - SRS Document**	**BA + SA**	**W4**	**SRS following IEEE 830**
2.1	Role-play interview stakeholders	BA	W4	Interview notes
2.2	Elicit & classify ≥16 FR + ≥7 NFR	BA + SA	W4	FR/NFR table with priority ratings
2.3	Write ≥10 Use Cases + User Stories	SA	W4	UC description + acceptance criteria
2.4	Complete SRS according to IEEE 830	BA + SA	W4	Submit D2
3.0	**D3 - System Modeling Package**	**SA**	**W5-W6**	**DFD + UML + ERD**
3.1	DFD Level 0 + Level 1	SA	W5	Balanced DFD diagrams
3.2	Class Diagram + ≥3 Sequence Diagrams	SA	W5-W6	UML diagrams in StarUML
3.3	ERD 3NF + Data Dictionary	SA + SD	W6	Logical + Physical ERD
3.4	Traceability matrix SRS ↔ Models	SA	W6	Submit D3
4.0	**D4 - System Design Document**	**SD**	**W7-W9**	**Complete SDD**
4.1	UI Wireframes ≥5 screens (Figma)	SD + QA	W7	Mobile app + web dashboard
4.2	Architecture + 2 alternatives + trade-off	SD	W8	Layered vs Microservices analysis
4.3	Physical data schema + security model	SD	W9	PostGIS schema + threat model

4.4	QA plan + test cases + ops plan	QA	W9	Submit D4
5.0	**D5 - Final Presentation & Defense**	**QA + All**	**W10**	**Slide + Demo + Q&A**

6. Project Timeline

Week	Phase	Main Activities	Deliverable	Milestone
W1	P1 - Business	Kickoff, domain selection, team formation, stakeholder screening.	**Topic proposal**	Groups + domains confirmed
W2	P1 - Business	Stakeholder map, AS-IS analysis, pain points.	**Business case draft**	Complete Stakeholder Map
W3	P1 - Business	Business case, WBS, Risk Register, Gantt chart, peer review.	**D1 Proposal**	SUBMIT D1 - end of W3
W4	P2 - Requirements	Interview, elicit FR/NFR, write UC + SRS.	**D2 SRS**	SUBMIT D2 - end of W4
W5	P3 - Modeling	PT1, DFD L0+L1, Class Diagram draft.	**PT1 + DFD draft**	Progress Test 1
W6	P3 - Modeling	Complete UML, ERD, and traceability matrix.	**D3 Models**	SUBMIT D3 - end of W6
W7	P4 - Design	UI wireframes Figma, persona, user journey.	**UI wireframes**	≥5 complete screens
W8	P4 - Design	PT2, Architecture pattern, tech stack.	**PT2 + Architecture**	Progress Test 2
W9	P4 - Design	Trade-off analysis, security model, QA plan, ops.	**D4 SDD**	SUBMIT D4 - end of W9
W10	P5 - Defense	PT3, mock defense, final presentation.	**D5 Defense**	FINAL PRESENTATION - W10

6.1. Gantt Chart – 10 Week Project Schedule

The chart below maps all deliverables and progress tests to their target weeks. Each color represents a distinct deliverable. Milestone weeks (D-submit weeks) are shaded green in the timeline table above.

Activity / Deliverable	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10
D1 - Project Proposal										
D2 - SRS (IEEE 830)										
PT1 - Progress Test 1										
D3 - System Modeling										
D4 - System Design										
PT2 - Progress Test 2										

PT3 - Progress Test 3										
D5 - Final Defense										

Legend: Blue = D1 Proposal | Green = D2 SRS | Orange = Progress Tests (PT1/PT2/PT3) | Yellow = D3 System Modeling | Purple = D4 System Design | Red = D5 Final Defense

7. Risk Register

#	Risk	Probability	Impact	Level	Mitigation
R1	**Scope creep extending too many modules**	High	High	Critical	Keep the out-of-scope frozen; each new feature requires PM approval before inclusion.
R2	**Lack of direct field data and stakeholder validation from agricultural cooperatives, farmers, and drone operators**	Medium	High	High	Use peer-reviewed studies, government regulations, verified industry reports, role-play interviews, and expert walkthroughs as preliminary proxy evidence. Clearly document the absence of direct field interviews as a study limitation and plan validation with real cooperative managers, farmers, or drone operators in a future pilot.
R3	**Member takes unexpected leave mid-term**	Low	High	High	Fully documented tasks on Notion; each deliverable has a designated backup member.
R4	**Inconsistency between SRS, UML, SDD**	High	High	Critical	Traceability matrix updated and reviewed after each deliverable submission.
R5	**Lack of time to complete D4 (Week 9)**	Medium	Medium	Medium	Begin architecture design from W7; do not allow work to accumulate into W9.
R6	**Legal basis of Decree 288/2025 is unclear, making compliance module design difficult**	Medium	Low	Medium	Scope compliance module at PDF report output level - not directly integrated with Operations Dept.
R7	**Regulation change risk - NĐ 288/2025 may be supplemented by additional decrees**	Low	Low	Low	Record assumption: 'Design is based on NĐ 288/2025 as of analysis date - requires review if new decrees are issued.'
R8	**Domain knowledge gap team lacks direct field experience in ĐBSCL**	Medium	Medium	Medium	Compensate via peer-reviewed literature (Nguyen et al. 2025), field reporting, and role-play stakeholder interviews in W4.
R9	**Simulated telemetry and mobile GPS may not accurately represent real field latency, packet loss,**	Medium	Medium	Medium	Use structured test scenarios and explicitly state that performance findings are design-level expectations requiring validation

	GPS drift, or connectivity conditions				with physical drones in a future pilot.
R10	**GPS-based operator and field tracking may create privacy and unauthorized-access risks**	Medium	Medium	Medium	Apply data minimization, task-bound location tracking, role-based access control, consent notification, access logging, and a defined retention policy.

8. Team Roles & Responsibilities

Member	Role	Main Deliverable	Notes
Dang Thanh Cong	Project Manager - planning, progress tracking, team coordination	Summary of all deliverables	Contact the instructor; maintain traceability matrix
Nguyen Thi Thanh Thao	Business Analyst - business analysis, stakeholder mapping, requirements elicitation	D1 Proposal, D2 SRS	Lead stakeholder interviews and FR/NFR elicitation
Le Van Minh	System Analyst - DFD, UML, ERD, and traceability models	D3 System Models	Use PlantUML for all diagrams
Tran Anh Khoa	System Designer - architecture, UI/UX Figma, data design, security	D4 SDD	Use Figma for wireframes; PostGIS for data schema
Dang Vo Thanh Hieu	Presenter / QA - slides, storytelling, quality control, Q&A preparation	D5 Presentation	Cross-check consistency across all deliverables

9. D1 Quality Checklist

Criteria	Completion	Reference
Stakeholders mapped (Primary / Secondary / Regulatory) with Power & Interest ratings	Complete	Section 3
Measurable problem statement with quantified impact (15–20% cost, 76,000 affected customers)	Complete	Section 1
Clear scope - In-scope (10 functions) and Out-of-scope (6 exclusions) explicitly listed	Complete	Section 1.3–1.4
Business case with specific data - 3-layer ecosystem, TAM/SAM/SOM, cost-benefit table, ROI	Complete	Section 2
WBS + Gantt chart + Timeline + Risk Register (≥5 risks with likelihood, impact, mitigation)	Complete	Sections 5, 6, 6.1, 7
AI Audit Log entries #01-08 included (W1-W3 phase coverage)	Complete	Section 10
Cost-benefit data source notes added for all quantitative figures	Complete	Section 2.3
SOM methodology disclaimer added	Complete	Section 2.2
R8 probability corrected (Low → Medium) with rationale note	Complete	Section 7
R9 + R10 added (simulated telemetry risk; GPS privacy risk)	Complete	Section 7

10. AI Artifact Audit Log - Entries #01-#08 (W1-W3)

This consolidated audit log covers all AI interactions for Phase 1 (Problem & Business). Entries #01–#03 are from W1, #04–#06 from W2, and #07–#08 from W3. The cumulative log continues in D2 (entries #09–#12) and will reach ≥25 entries by end of W10.

v1.6 refinement (2026-06-25) – Decision & Human Delta: During internal review the team applied six cross-cutting corrections to the entries above without altering the original AI prompts or tools. (1) The 50-drone figure was reclassified from a hard limit to a configurable reference workload / validation baseline. (2) An inconsistency was identified between the SaaS subscription price and the provider-side infrastructure cost; the ROI was recomputed from the cooperative's perspective (~15M VND/season subscription vs ~21.9M VND gross savings, net ~6.9M VND, preliminary ROI ≈ 46%, subject to field validation). (3) Safety claims were reduced to information-level hazard alerting because the platform does not directly control drones, firmware, or autopilot. (4) Simulated and mobile-GPS telemetry were distinguished from physical-drone integration, with native MAVLink moved to post-MVP future work. (5) Absolute market-gap claims were replaced with evidence-bounded statements. (6) Published-literature evidence was distinguished from scenario assumptions throughout the business case. (7) Source citation for Long An 110kV incident corrected from Dan Viet (2024) to Tuoi Tre (2024) following source verification. (8) The Operations Dept. stakeholder note's claim that Cục Tác chiến's flight-permit role is governed by Decree 288/2025/ND-CP was found to overstate what the cited source supports. Vietnam Naval News [ref. 6, year corrected 2024→2025] confirms only that flight-permit dossiers are routed to Cục Tác chiến; the decree itself (Điều 21, Khoản 2) assigns authority to the Ministry of National Defence and Ministry of Public Security. The note has been reworded to separate the verified claim from the decree-linkage inference. (9) The TAM figure (~4,000 drones / 1.5M ha) lacked a supporting source — neither Vietnam News (2024) nor Nguyen et al. (2025) contains this figure. Replaced with >3,000 drones / ~1.5M ha, sourced to VnExpress (2025) citing Markets & Data, which directly states this figure.

No.	Week	DTC Component	AI Tool	Prompt (Full Context)	AI Output (Summary)	Decision & Human Delta / Verification
01	W1 / P1	Problem & Business Decomposition	Claude	"We are developing an IS project for ITA301 on Agricultural UAVs in the Mekong Delta. Suggest 3 distinct sub-domain directions, stating the primary business problem and implementation complexity for each."	3 directions: (1) Fleet Management generic tracking; (2) Precision Spraying Platform localized scheduling; (3) Marketplace hourly drone rentals. All rated 3 star complexity.	**SELECTED (2) Precision Spraying Platform: Empirical data: Nguyen et al. 2025, N=940. Pain points validated: HTX uses Zalo/Excel. Platform layer identified as a market gap in the ecosystem based on reviewed evidence.** **REJECTED (3): Scope too broad, no IS depth.** **REJECTED (1): Overlaps with another team's confirmed topic.** Verification: Ecosystem gap confirmed - Lao Dong (2025) confirms digital scheduling problem remains unresolved in ĐBSCL cooperatives.
02	W1 / P1	Pattern Recognition & Comparison	ChatGPT	"Compare Agricultural UAV, Delivery UAV, and Surveillance UAV across: (a) stakeholder complexity, (b) regulatory burden in Vietnam, and (c) data availability for an IS academic project."	AI concluded Delivery UAV is 'easiest' due to lower perceived regulation. Agricultural UAV ranked medium complexity.	**AI RANKING REJECTED incorrect evaluation: Agricultural UAV has a more structured legal framework (Decree 288/2025/ND-CP + Plant Protection Dept) than Delivery UAV.** **Maintained Agricultural UAV: (a) peer-reviewed research available (Nguyen 2025), (b) verifiable field incidents, (c) richer stakeholder map for IS analysis.** Verification: Decree 288/2025/ND-CP cross-checked via thuvienphapluat.vn confirms specific drone ID and pilot certification logging requirements.

03	W1 / P1	Decomposition - Ecosystem Mapping	Claude	"For the AgriUAV Platform (ITA301, Mekong Delta): decompose the current Agricultural UAV ecosystem into structural layers. Identify which layers are occupied and which represent market gaps for an IS solution."	AI proposed a 2-layer split: Hardware/Device layer and Application/User layer. Suggested the gap lies in 'application integration.' No distinct Service Layer or Platform Layer identified.	**REJECTED 2-layer model oversimplified and inaccurate.** **Team defined 3-Layer Model: Hardware Layer (DJI/XAG occupied), Service Layer (HTX fleets occupied, manual), Platform Layer (digital IS coordination identified as the market gap based on reviewed evidence).** Verification: 3-layer model validated by Thanh Nien (June 2025): 1,215 active drones across 3 provinces, with no platform-layer management software described in the source.
04	W2 / P1	Decomposition - Stakeholder Mapping	Claude	"Project: AgriUAV Platform for drone spraying in Mekong Delta. List all stakeholders, classify as Primary/Secondary/Regulatory, and outline core roles and expectations for each."	Suggested 6 general groups: Farmers, HTX, Pilots, Drone Manufacturers (DJI), Software Developers, and Local Government totaling ~12 stakeholders.	**5 core stakeholders accepted.** **Removed 'Software Developer Team' this is the project team, not an external stakeholder.** **SUPPLEMENTED 2 critical actors AI omitted: Operations Dept. / General Staff (sole authority for UAV flight permits under Decree 288/2025); Pesticide Retailers (essential for inventory/compliance integration).** **Split generic 'Government' into 2 specific regulatory bodies: Plant Protection Dept. + Operations Dept.** Verification: Operations Dept. authority confirmed via Vietnam Naval News (2025).
05	W2 / P1	Decomposition - AS-IS Process Analysis	Claude	"Decompose the AS-IS business process of an agricultural cooperative in Vietnam executing drone spraying services. List each step, the actor responsible, and pain points at each stage. Context: currently managed via Zalo/Excel."	Proposed a basic 4-step generic workflow: Intake → Dispatch → Execution → Reporting. High-level pain points only: lack of tracking, slow	**Accepted 4-step structure as baseline.** **Expanded to 5 steps added 'Monitoring & Control' as a distinct phase (AI merged this into Execution).** **Localized pain points: Phase 3 risk of inaccurate or**

					report compilation.	**unverifiable flight-hour and chemical-usage records (AI missed this risk entirely); Phase 4 Long An 110kV power line incident (Tuoi Tre 2024) as evidence of the need for hazard visibility and incident logging; Phase 5 linked compliance failure directly to Decree 288/2025/ND-CP.** Verification: 110kV incident: Tuoi Tre (2024) drone collision in Long An Province, 76,000 customer blackout.
06	W2 / P1	Process / Algorithm - Business Case	Claude	"Provide an industry-standard business case structure for an IS research project, including a preliminary ROI template and a formal problem statement formulation suitable for a university-level IS analysis project."	Standard 5-part structure (Context → Problem → Impact → Solution → Scope) and a generic theoretical ROI formula: (cost saved / investment) × 100%.	**Adopted structural blueprint as foundation.** **Applied Mekong Delta data localization: 15–20% operational cost figure from Nguyen et al. (2025) N=940; 1,215 drones across 3 provinces from Thanh Nien June 2025.** **Added HTX Cost-Benefit table (not in AI output): pesticide –30%, fuel –20%, labor –75% as scenario assumptions = ~21.9M VND gross savings; corrected to compare against the HTX subscription fee of ~15M VND/season (not the ~3.6M VND provider-side infrastructure cost), giving a preliminary ROI ≈ 46% subject to field validation.** Note: absolute VND figures in cost-benefit table are team derivations from published operational benchmarks see methodology notes in Section 2.3. Verification: 15–20% cost: Nguyen et al. (2025). Province drone counts: Thanh Nien (June 2025).

						[Correction — 2026-06-25, D1 v1.6] The 15–20% operational-cost-inflation figure is a team estimate (planning assumption), not a value reported in Nguyen et al. (2025). That paper (RWAE Journal; N=940) is a Probit analysis of UAV-adoption determinants and supplies only the sample size and the average farmer age (~50.88 yrs, Table 1) cited elsewhere in this report; it does not measure operational-cost inflation. The (Nguyen et al., 2025) citation has accordingly been removed from the 15–20% figure throughout the report body and the figure relabelled as a team estimate. The original entry above is retained as a process record.
07	W3 / P1	Pattern Recognition - Business Case Comparison	Claude	"Compare the business case of an Agricultural UAV platform vs. a Delivery UAV platform in Vietnam: market size, competition level, scalability, and suitability for IS research."	AI concluded Delivery UAV has 'a larger market' because logistics is booming. Agricultural UAV was rated lower on 'tech sophistication.'	**COUNTER-ARGUMENT AGAINST AI BIAS: AI conflated 'market size = revenue potential' with 'research quality.' ITA301 evaluates domain richness, not revenue.** **Agricultural UAV is more suitable: (1) peer-reviewed paper exists (Nguyen et al. 2025, RWAE Journal) no equivalent for Delivery UAV in Vietnam; (2) real field data available by province from Vietnamese press; (3) richer stakeholder map; (4) platform-layer gap supportable via 3-layer ecosystem decomposition. Team maintained Agricultural UAV selection. AI bias pattern recognized and documented.** Verification: Pattern recognition team identified AI bias toward 'market =

						revenue' framing vs. ITA301 standard of 'domain richness = research quality.'
08	W3 / P1	Decomposition - WBS & Risk Register	Claude	"Propose a Risk Register for a 10-week IT project analyzing agricultural drone spraying in ĐBSCL. List 8–10 risks with likelihood (H/M/L), impact, and mitigation strategy."	AI listed 10 risks: scope creep, timeline, team conflict, data quality, tool learning curve, stakeholder availability, budget, integration complexity, security breach, deployment failure.	**Accepted 5 generic risks (scope creep, timeline, team conflict, stakeholder availability, data quality).** **REMOVED 'deployment failure' ITA301 does not require actual deployment.** **REMOVED 'budget' not applicable to a university project.** **SUPPLEMENTED 4 UAV-domain-specific risks AI missed: Regulation change risk (NĐ 288/2025 newly effective, supplementary decrees possible); Domain knowledge gap (team lacks direct field experience probability set to Medium, corrected from Low); Literature availability; Stakeholder interview difficulty.** Verification: NĐ 288/2025 confirmed effective via VnBusiness and Government E-Portal.

11. Verified Academic & Regulatory References

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Note: This article discusses the regulatory gap that existed before Decree 288/2025/ND-CP took effect and confirms the Operations Department's role in processing flight-permit dossiers; it does not itself reference Decree 288/2025/ND-CP.

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